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(a) Since the system is in equilibrium, net force = 0

Upthrust on cube = Weight of two cubes

$$\rho_{fluid} r^2 (2r - h) g = (y - x) g$$
$$\rho_{fluid} = \frac{(y-x)}{r^2(2r-h)}$$

(b) - The cube that remains in the fluid is at rest at the point when the top cube is removed. Since the upthrust at that instant is larger than the weight of the cube, the net force is upwards so the velocity starts to increase in the upward direction.

- As the cube moves upwards, the upthrust decreases till it is the same magnitude as the weight of the cube at equilibrium position where the magnitude of the velocity is maximum.

- The upthrust decreases further resulting in the net force being downwards as it moves further up. This results in the velocity decreasing as it reaches its highest position where the velocity comes to zero.

- This shows that the cube that remains in the fluid is in oscillating motion.

(c) (i) Height starts from lowest point (1.30 cm) to highest point (2.00 cm) with equilibrium position at 1.65 cm. Height at $t=0$ is 1.30 cm and increasing. Reaches 2.00 cm after half a period.

Sinusoidal

Label period and starts from lowest point

(ii) Kinetic energy graph is sinusoidal. 2 cycles shown within a period. KE is 0 at 0, $T/2$ and T . KE is max at $T/4$ and $3T/4$.

Sinusoidal

2 maxima within one period and starts from zero

